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GOVERNMENT OF KARNATAKA
DEPARTMENT OF COLLEGIATE AND TECHNICAL EDUCATION
GOVERNMENT ENGINEERING COLLEGE NARGUND-582207



A Major Project
REPORT ON
“NEXT-GEN PATIENT CARING SYSTEM”

Submitted in partial fulfillment of the requirement for the degree of
Bachelor of Engineering
in
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

SUBMITTED BY:

- | | |
|---------------------------------------|------------|
| 1. Aishwarya Basanagouda Murigennavar | 2NR22AD001 |
| 2. Jyothi M H | 2NR22AD011 |
| 3. Sameena Kausar Magami | 2NR22AD028 |
| 4. Sanjana M H | 2NR23AD413 |

Under The Guidance of
Prof. Shweta Ganiger

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE
GOVERNMENT ENGINEERING COLLEGE NARGUND-582207

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DEPARTMENT OF COLLEGIATE AND TECHNICAL EDUCATION
GOVERNMENT ENGINEERING COLLEGE NARGUND-582207

(Affiliated To Visvesvaraya Technological University)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE 2025-26

Certificate

Certify that the project work entitled “**Next-Gen Patient Caring System**” carried out by, **Aishwarya Basanagouda Murigennavar (2NR22AD001), Jyothi M H (2NR22AD011), Sameena Kausar Magami (2NR22AD028), Sanjana M H (2NR23AD413)** are bonafide students of 7th semester in partial fulfillment for the award of **Bachelor of Artificial intelligence and Data Science of The Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections or suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the aid degree.

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Guide

Signature of the
Project Coordinator

Signature of the
HOD

Signature of the
Principal

External Viva

Name of the Examiners

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1)

.....

2)

.....

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This is to declare that the Dissertation work entitled " **Next-Gen Patient Caring System** " is a bonafide work carried out by us at Government Engineering College, NARGUND in partial fulfillment of the requirements for the award of the degree of BACHELOR OF ENGINEERING in ARTIFICIAL INTELLIGENCE & DATA SCIENCE of VISVESWARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI under the guidance of **Prof. Shweta Ganiger**. Further it is declared to the best of our knowledge the work reported here in, does not form part of any other thesis or dissertation on the basis of which any other candidate was conferred a degree or award on earlier occasion.

Place: Nargund

Date:

1. Aishwarya Basanagouda Murigennavar	2NR22AD001
2. Jyothi M H	2NR22AD011
3. Sameena Kausar Magami	2NR22AD028
4. Sanjana M H	2NR23AD413

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|---------------------------------------|------------|
| 1. Aishwarya Basanagouda Murigennavar | 2NR22AD001 |
| 2. Jyothi M H | 2NR22AD011 |
| 3. Sameena Kausar Magami | 2NR22AD028 |
| 4. Sanjana M H | 2NR23AD413 |

ABSTRACT

The Next Generation Food Feeding System is an innovative solution designed to automate and optimize the delivery of nutrition to patients in hospitals, rehabilitation centers, and elderly care facilities. Traditional feeding methods often rely heavily on manual labor, which can lead to inconsistencies, delays, and increased workloads for healthcare staff. This system integrates advanced technologies to ensure accurate, timely, and hygienic feeding support for patients who require assistance. At the core of this system are intelligent sensors, automated controls, and adaptive mechanisms that monitor patient needs in real time. The system can adjust feeding speed, portion size, and timing based on patient conditions such as swallowing ability, nutritional requirements, and medical restrictions. By incorporating features like smart scheduling and data logging, the solution enhances patient safety and maintains strict adherence to prescribed dietary plans. Overall, the next generation patient feeding system aims to enhance patient comfort, support clinical efficiency, and reduce manual workload. By combining automation, intelligence, and precision, it provides a reliable and patient-centric approach to nutrition management in modern healthcare environments.

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CHAPTER – 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Healthcare systems across the world are continuously evolving with the integration of modern technologies such as automation, artificial intelligence, computer vision, and the Internet of Things (IoT). One of the major challenges faced in hospitals, rehabilitation centers, and home-care environments is providing continuous and personalized care to patients who have limited mobility or physical disabilities. Among daily care activities, feeding and hydration are critical yet often require constant human assistance. This dependence increases the workload on caregivers and may lead to delays, discomfort, and reduced independence for patients. To address these challenges, the Next Generation Patient Caring System is proposed as an innovative and assistive solution aimed at improving patient comfort, safety, and autonomy.

The Next Generation Patient Caring System is designed to assist patients who are unable to feed themselves due to physical impairments, neurological conditions, post-surgical recovery, or age-related limitations. The system focuses on hands-free feeding, particularly liquid intake, by automatically detecting the patient's mouth movement and delivering liquid in a controlled and safe manner. By leveraging computer vision techniques and embedded system technologies, the proposed model reduces the need for continuous manual intervention while ensuring timely and accurate feeding.

In traditional healthcare setups, patient feeding is primarily dependent on caregivers or nursing staff. While caregivers play an essential role, continuous manual assistance can be physically demanding, time-consuming, and prone to human error. In situations where a patient requires frequent feeding or hydration, delays can lead to discomfort, dehydration, or health complications. Moreover, during emergencies or staff shortages, patients may not receive timely care. The proposed system aims to overcome these limitations by introducing an automated feeding mechanism that operates based on real-time detection of patient intent.

The core concept of the Next Generation Patient Caring System revolves around mouth detection using a camera and computer vision techniques. When the system detects that the patient's mouth is open, it interprets this action as a feeding request.

The detected signal is processed by a microcontroller, which then activates a motor-driven water pump to dispense liquid in a controlled manner. Additional safety measures, such as sensor feedback, ensure that the liquid is delivered only when the patient is in the correct position and ready to receive it. This approach minimizes spillage, choking risks, and unintended activation.

The system is built using cost-effective and easily available hardware components such as an Arduino Uno microcontroller, motor shield, submersible water pump, camera module, and a 7.4V DC battery. Software tools such as OpenCV and Mediapipe are used for real-time face and mouth detection, while serial communication enables seamless interaction between the laptop-based vision system and the microcontroller. This combination of hardware and software allows the system to operate efficiently in real-world environments without requiring complex or expensive infrastructure.

One of the key advantages of the proposed system is its adaptability to different patient needs. Since the feeding process is initiated based on mouth movement rather than manual switches or buttons, it is highly suitable for patients with severe motor impairments. The system promotes independence by allowing patients to control their feeding process naturally and intuitively. Furthermore, the modular design makes it easy to upgrade or extend the system in the future by incorporating advanced features such as voice commands, mobile app monitoring, or cloud-based health data analysis.

Safety and reliability are critical aspects of any healthcare-related system. The Next Generation Patient Caring System incorporates multiple safety protocols to ensure patient well-being. The feeding mechanism operates only when predefined conditions are satisfied, and the system can immediately stop dispensing liquid if the mouth is closed or if abnormal conditions are detected. This ensures controlled feeding and reduces the risk of overfeeding or aspiration. Additionally, the use of low-voltage DC power enhances electrical safety, making the system suitable for prolonged use in healthcare environments.

1.2 APPLICATIONS

- The Next Generation Patient Caring System has wide-ranging applications across modern healthcare and assisted-living environments. In hospitals, the system can be deployed to support patients in general wards, intensive care units, and long-term care units by providing automated and timely liquid feeding, thereby reducing the dependency on nursing staff for routine hydration tasks. This not only improves

patient comfort but also allows healthcare professionals to focus on critical medical procedures and monitoring.

- In home-care settings, the system plays a vital role in assisting elderly individuals, patients with physical disabilities, and those recovering from surgeries or strokes. By enabling hands-free feeding through mouth detection, the system helps patients maintain independence and dignity while ensuring adequate hydration. Rehabilitation center can utilize the system as a supportive tool for patients undergoing motor or neurological recovery, allowing them to gradually adapt to independent feeding in a controlled and safe manner.
- The system is also highly beneficial in elderly care facilities, where residents often experience reduced mobility, tremors, or coordination issues. Automated feeding minimizes spillage and choking risks, enhancing safety and comfort. Furthermore, in smart hospitals and IoT-enabled healthcare environments, the system can be integrated with monitoring platforms to track usage patterns, generate alerts, and support data-driven patient care. Overall, the Next Generation Patient Caring System serves as an effective solution for improving quality of life, optimizing healthcare resources, and advancing intelligent patient care technologies.

1.3 ADVANTAGES

1.3.1 Hands-Free Operation

Enables patients to receive liquid without using their hands, making it ideal for bedridden or physically disabled individuals.

1.3.2 Improves Patient Independence

Allows patients to control their feeding process naturally through mouth detection, enhancing dignity and confidence.

1.3.3 Reduces Caregiver Workload

Minimizes the need for continuous manual assistance, allowing nurses and caregivers to focus on critical tasks.

1.3.4 Safe and Controlled Feeding

Dispenses liquid only when required, reducing risks of spillage, choking, and overfeeding.

1.3.5 Cost-Effective Solution

Built using low-cost, readily available components such as Arduino and DC pumps.

1.3.6 Energy Efficient

Operates on low-voltage DC power, ensuring minimal power consumption and safe usage.

1.3.7 Easy to Use and Maintain

Simple design with minimal moving parts results in low maintenance requirements.

1.3.8 Portable and Battery Operated

Can function during power outages and be easily moved between locations.

1.3.9 Scalable and Upgradable:

Supports future enhancements such as IoT monitoring, mobile app integration, and voice control.

1.3.10 Supports Smart Healthcare Systems

Aligns with modern healthcare automation and IoT-based patient care initiatives.

1.4 DISADVANTAGES**1.4.1 Limited to Liquid Feeding**

The system is mainly designed for liquids and cannot handle solid or semi-solid food.

1.4.2 Dependency on Camera Accuracy

Mouth detection performance may reduce under poor lighting conditions or camera misalignment.

1.4.3 Initial Setup Requirement

Proper positioning of the camera and pump setup is required for accurate operation.

1.4.4 Battery Dependence

Continuous operation depends on battery charge and requires regular recharging.

1.4.5 Limited Adaptability to Patient Movement

Sudden head or body movements may affect detection accuracy.

1.5 SCOPE OF THE PROJECT

The scope of the Next Generation Patient Caring System is to design and implement an automated, hands-free liquid feeding solution that assists patients with limited mobility in hospitals, care centers, and home-care environments. The project focuses on developing a reliable system that detects mouth opening using computer vision techniques and activates a motor-controlled water pump to dispense liquid safely and efficiently.

The scope includes the integration of hardware components such as an Arduino Uno microcontroller, motor shield, submersible water pump, camera module, and a 7.4V battery, along with software tools like OpenCV and Mediapipe for real-time mouth detection. The system aims to provide controlled liquid dispensing while ensuring patient safety through basic sensor feedback and control logic.

The project scope also allows for future enhancements such as IoT connectivity for remote monitoring, advanced safety mechanisms, and adaptive feeding control based on patient behavior. However, the current scope is limited to liquid feeding assistance and does not include solid food handling or advanced medical diagnostics. Overall, this project demonstrates the application of embedded systems and smart technologies in healthcare automation, serving as a foundation for future intelligent patient care solutions.

1.6 PROBLEMS IN TRADITIONAL PATIENT CARING SYSTEM

Traditional patient caring systems mainly depend on manual work by nurses, caretakers, and family members. While this approach has been followed for many years, it faces several serious limitations in today's healthcare environment.

1.6.1 Heavy Dependence on Human Labor

In traditional systems:

- Patients require continuous physical presence of nurses or caretakers.
- Tasks such as feeding, medicine delivery, monitoring vitals, and movement assistance are done manually.

Problems:

- Human fatigue and workload increase errors.
- Limited staff cannot monitor multiple patients continuously.
- Care quality depends heavily on individual experience.

1.6.2 Lack of Continuous Monitoring

- Most traditional systems:
- Check patient condition at fixed intervals (hourly or shift-based).
- Use manual observation rather than real-time monitoring.

Problems:

- Sudden health changes may go unnoticed.
- Critical conditions (breathing issues, unconsciousness, falls) are detected late.
- Delayed response can lead to serious complications.

1.6.3 High Risk of Human Error

Manual systems involve:

- Handwritten records
- Verbal instructions
- Manual medication schedules

Problems:

- Wrong dosage or missed medication
- Incorrect feeding time
- Misinterpretation of patient condition
- These errors directly affect patient safety.

1.6.4 Inadequate Care for Elderly and Disabled Patients

Traditional care struggles with:

- Patients with paralysis
- Elderly patients living alone
- Patients with speech or movement disabilities

Problems:

- Patients cannot express needs clearly.
- Dependence on others reduces independence.
- Delayed help during emergencies.

1.6.5 Increased Cost of Long-Term Care

Manual patient care requires:

- More nurses and caretakers
- Longer hospital stays
- Repeated routine checks

Problems:

- High operational cost for hospitals
- Financial burden on families
- Inefficient use of medical staff

1.6.6 Poor Data Management and Documentation

Traditional systems rely on:

- Paper records
- Manual updates

Problems:

- Data loss or damage
- Difficult to analyze patient history
- No real-time analytics or prediction

1.6.7 Limited Emergency Response Capability

In emergency situations:

- Caretaker must first notice the issue.
- Manual calling or reporting is required.

Problems:

- Slow response time
- No automatic alert system
- Higher risk during night hours or understaffed shifts

1.7 NEED FOR AUTOMATION IN PATIENT CARING SYSTEM

Automation in patient caring systems uses technology such as sensors, microcontrollers, AI, computer vision, and IoT to provide continuous, accurate, and intelligent care.

1.7.1 Continuous Real-Time Monitoring

Automated systems:

- Monitor vital signs continuously.
- Track patient posture, movement, and activity.

Benefits:

- Early detection of abnormalities.
- Immediate alerts to caregivers or doctors.
- Reduced risk of unnoticed emergencies.

1.7.2 Reduction of Human Workload

Automation handles:

- Routine tasks (feeding, medicine reminders).
- Monitoring and data logging.

Benefits:

- Nurses can focus on critical patients.
- Reduced physical and mental stress.
- Improved efficiency of healthcare staff.

1.7.3 Improved Patient Safety and Accuracy

Automated systems:

- Follow programmed schedules.
- Use sensors and algorithms for precision.

Benefits:

- Eliminates missed medication.
- Ensures correct feeding quantity and timing.
- Reduces medical errors.

1.7.4 Support for Elderly, Disabled, and Critical Patients

Automation enables:

- Gesture recognition (using OpenCV & MediaPipe).
- Voice or touch-based alerts.
- Motorized feeding and movement assistance.

Benefits:

- Increases patient independence.
- Improves quality of life.

- Provides dignity and comfort.

1.7.5 Faster Emergency Detection and Alerts

Automated systems can:

- Detect falls, unconsciousness, or abnormal vitals.
- Send instant alerts via alarms or messages.

Benefits:

- Quick medical intervention.
- Life-saving response time.
- 24/7 monitoring without fatigue.

1.7.6 Better Data Collection and Analysis

Automated systems:

- Store data digitally.
- Maintain patient history.

Benefits:

- Easy analysis and reporting.
- Helps doctors make better decisions.
- Supports predictive healthcare.

1.7.7 Cost-Effective Long-Term Solution

Though initial setup cost is high:

- Operational cost reduces over time.
- Fewer staff needed for routine tasks.

Benefits:

- Affordable long-term care.
- Scalable for hospitals and home care.
- Efficient resource utilization.

1.7.8 Integration with Smart Healthcare Technologies

Automation supports:

- IoT connectivity.
- AI-based analysis.
- Remote monitoring.

Benefits:

- Doctors can monitor patients remotely.
- Enables telemedicine.
- Future-ready healthcare system.

CHAPTER – 2

OBJECTIVES AND MOTIVATION

2.1 MOTIVATION

The development of the Next Generation Patient Caring System is strongly motivated by the growing need for intelligent, automated healthcare solutions that can support patients with limited mobility and reduce dependence on constant human assistance. In hospitals, rehabilitation centers, and home-care environments, many patients face difficulty in performing basic daily activities such as drinking water due to paralysis, muscle weakness, post-surgical conditions, or age-related disorders. These challenges often result in discomfort, dehydration, and loss of independence.

Another major motivation is the increasing burden on caregivers and healthcare professionals. Nurses and attendants are required to assist multiple patients simultaneously, making it difficult to provide continuous and personalized care. Repetitive tasks such as feeding and hydration consume significant time and physical effort. An automated patient caring system can effectively reduce this workload, allowing caregivers to focus on more critical medical responsibilities.

The system is also motivated by the need to improve patient dignity and psychological well-being. Dependence on others for basic needs can cause emotional stress, embarrassment, and loss of self-confidence among patients. By enabling hands-free and intent-based feeding through mouth detection, the system empowers patients to control their own hydration process in a natural and non-invasive manner.

From a technological perspective, recent advancements in computer vision, embedded systems, and IoT have made it feasible to design affordable and reliable assistive devices. The integration of camera-based mouth detection with microcontroller-controlled liquid dispensing demonstrates how modern technologies can be applied to solve real-world healthcare problems. The motivation is to bridge the gap between advanced research and practical, low-cost healthcare. Additionally, the system addresses safety concerns associated with manual feeding, such as choking, spillage, and overfeeding. By incorporating controlled dispensing and sensor feedback, the system ensures accurate and safe operation. This is particularly important for patients who are unable to communicate verbally or have limited

control over their movements. Overall, the motivation behind this project is to create a smart, accessible, and scalable patient care solution that enhances quality of life, improves healthcare efficiency, and supports the vision of future smart hospitals and connected healthcare systems.

2.2 PROBLEM STATEMENT

Many individuals with physical disabilities, age-related impairments, or temporary health conditions struggle with the basic act of feeding themselves. Relying on caregivers can lead to a loss of independence and dignity. There is a need for an automated, intelligent, and hygienic system to assist such individuals in feeding themselves. For individuals with physical disabilities, paralysis, elderly health conditions, or temporary injuries, eating food independently often becomes a major challenge. Dependence on caregivers for basic activities like eating can impact a person's dignity, self-confidence, and overall quality of life. While caregivers provide necessary assistance, constant dependency is not always feasible due to limited availability, cost, or emotional strain.

2.3 SOLUTION TO THE PROBLEM

The Smart Automated Food Feeding System provides an intelligent and hygienic solution to assist elderly people, disabled individuals, and hospital patients who face difficulty in feeding themselves. The proposed solution is a sensor-based automatic liquid feeding system designed to assist patients who are unable to feed themselves. The system uses a mouth detection sensor—such as an infrared or Camera sensor—to identify when the patient's mouth is open. Once detected, the Arduino activates a small pump to dispense a controlled amount of liquid through a hygienic feeding tube.

When the mouth closes, the sensor signal changes and the pump automatically stops, preventing overflow or choking. This hands-free mechanism eliminates the need for manual feeding, ensuring comfort and safety for bedridden or physically challenged patients. The system also includes safety features such as an emergency stop switch, time-based flow control, and visual indicators to alert caregivers of the system status. By combining simple electronics, automation, and human-centric design, this model offers an affordable and reliable solution for improving patient independence and caregiver efficiency in healthcare environments.

2.4 OBJECTIVES

The primary objective of the Next Generation Patient Caring System is to design and develop an automated, hands-free feeding solution that enhances patient comfort, safety, and independence. The specific objectives of the system are as follows:

- To develop an automated feeding system that allows patients to consume liquids without using their hands or requiring constant caregiver assistance.
- To detect mouth opening accurately using camera-based computer vision techniques to identify the patient's intent for feeding.
- To control liquid dispensing safely and precisely using a Arduino and motor-driven water pump.
- To minimize caregiver workload by automating routine feeding tasks in hospitals, care centers, and home-care environments.
- To ensure patient safety by incorporating sensor feedback and control logic to prevent overfeeding or accidental liquid flow.
- To design a cost-effective and energy-efficient system using readily available hardware components.
- To promote patient independence and dignity, especially for individuals with physical disabilities, elderly patients, or those recovering from surgery.
- To promote user independence and dignity by enabling individuals to feed themselves with minimal caregiver assistance.

CHAPTER – 3

LITERATURE SURVEY

A literature survey is a systematic study and summary of existing research, technologies, and published works related to your project topic. It shows that you understand what work has already been done and how your project fits into or improves upon existing systems.

[1]. “Kumar et al. (2018) presented an Internet of Things (IoT)-based smart patient monitoring system”

Designed to enhance healthcare delivery through continuous and real-time monitoring of vital physiological parameters. The primary objective of the proposed system was to reduce the dependency on frequent hospital visits while ensuring timely medical intervention through remote patient supervision. The study focused on monitoring critical health parameters such as heart rate, body temperature, and blood pressure, which are essential indicators of a patient's overall health condition. The system architecture comprised multiple biomedical sensors interconnected through an IoT framework. These sensors were strategically placed on the patient's body to capture real-time physiological data continuously. The collected data was transmitted wirelessly to a cloud-based platform, enabling centralized data storage and processing. Healthcare professionals could access this information remotely through web or mobile interfaces, allowing them to monitor patients' health conditions regardless of geographical location. This feature was particularly beneficial for elderly patients, individuals with chronic illnesses, and patients residing in remote or rural areas.

[2]. “Shakil et al. (2020) proposed an advanced AI-enabled remote patient care system”

Aimed at enhancing healthcare services through intelligent data analysis and early disease detection. The primary goal of the study was to shift healthcare delivery from a reactive approach to a proactive and predictive model by integrating artificial intelligence and machine learning techniques into remote patient monitoring frameworks. The system focused on identifying potential health risks and medical abnormalities at an early stage, thereby enabling timely intervention and improving patient outcomes. The proposed system utilized machine learning algorithms to analyze both historical patient records and real-time physiological data collected from various health monitoring devices. These datasets included

vital signs, clinical history, and behavioral health indicators. By processing this large volume of data, the system was able to detect hidden patterns, correlations, and anomalies that might not be easily identified through conventional monitoring methods. This intelligent analysis significantly enhanced diagnostic accuracy and supported clinicians in making informed decisions.

[3]. “Li et al. (2023) proposed an advanced intelligent patient caring system”

That integrates Internet of Things (IoT) technologies with deep learning techniques to deliver personalized, predictive, and data-driven healthcare services. The main objective of the study was to enhance traditional patient care systems by incorporating intelligent decision-making capabilities that can analyze complex medical data, predict health risks, and provide customized healthcare recommendations. This approach represents a significant step toward next-generation smart healthcare systems focused on individualized patient treatment and proactive medical intervention. The proposed system architecture consisted of IoT-enabled medical sensors used to continuously collect real-time patient health data, including vital signs, physiological parameters, and behavioral information. These data were transmitted securely to centralized processing units, where deep learning models were applied to extract meaningful insights. The deep learning algorithms were trained on large-scale patient datasets to recognize complex patterns, correlations, and temporal trends that are difficult to capture using conventional rule-based or statistical methods.

[4]. “Zhao et al. (2023) introduced an edge-AI enabled next-generation patient caring system”

Aimed at improving real-time health monitoring, rapid decision-making, and emergency response in modern healthcare environments. The primary motivation of the study was to overcome the limitations of traditional cloud-centric healthcare systems, particularly issues related to latency, network dependency, and delayed response during critical medical situations. By shifting data processing and intelligent analysis closer to the data source, the proposed system sought to enhance system responsiveness and patient safety. The system architecture was based on the integration of IoT-enabled medical sensors with edge computing devices equipped with artificial intelligence capabilities. Patient health data, such as vital signs and physiological parameters, were collected continuously through sensors and processed locally at the edge nodes. Instead of transmitting all raw data to centralized cloud servers, the edge-AI framework performed real-time data analysis, anomaly detection, and

decision-making directly at the edge level. This approach significantly reduced data transmission delays and minimized dependence on continuous internet connectivity.

[5]. “Singh and Kaur (2022) proposed an artificial intelligence–based health risk prediction system”

Aimed at enhancing proactive healthcare management through intelligent analysis of patient data. The primary objective of the study was to improve early detection of potential medical emergencies by leveraging machine learning algorithms to identify abnormal patterns and trends in both historical and real-time patient health records. The proposed system sought to overcome the limitations of conventional healthcare monitoring approaches that rely primarily on fixed threshold values and manual observation. The system architecture was designed to collect and process diverse types of patient data, including vital signs, clinical history, and continuous monitoring information. Machine learning models were employed to analyze these datasets and extract meaningful patterns associated with health deterioration or emergency conditions. By learning from historical data and adapting to real-time inputs, the system was capable of predicting potential health risks before they became critical, thereby enabling timely medical intervention and improved patient outcomes.

CHAPTER – 4

EXISTING SYSTEM

The existing food feeding system is largely based on traditional manual and semi-automated approaches where feeding is performed by caregivers, nurses, or family members, making the patient completely dependent on human support. In hospitals, rehabilitation centers, and home-care settings, this method requires continuous attention and physical presence, which increases workload, time consumption, and the possibility of human error. Some mechanical or motor-based feeding devices are available, but these systems' function using predefined time intervals or simple switch-based controls and do not adapt to the patient's real-time condition. They are unable to detect important parameters such as mouth opening, facial expressions, swallowing readiness, or patient comfort. As a result, food may be dispensed even when the patient is not ready, leading to food spillage, wastage, or serious safety issues like choking and aspiration.

Conventional tube-feeding systems also lack intelligence and feedback mechanisms. They continuously supply liquid food at a constant rate without monitoring user response, which can cause discomfort and digestive complications. These systems do not provide personalization based on individual feeding speed or behavior. Moreover, most existing systems do not integrate advanced technologies such as computer vision, sensors, or artificial intelligence to analyze user actions or make intelligent decisions. They are often bulky, expensive, and difficult to operate independently, limiting their usability in home environments. The absence of automation, adaptability, and safety monitoring in the existing food feeding systems makes them inefficient and unreliable, especially for long-term care patients, clearly indicating the need for a smarter, safer, and more user-friendly feeding solution.

CHAPTER – 5

REQUIREMENT SPECIFICATION

5.1 FUNCTIONAL REQUIREMENTS

- The system shall capture real-time video input of the patient using a camera.
- The system shall detect the patient's face before performing mouth detection.
- The system shall accurately identify mouth open and mouth closed states.
- The system shall initiate liquid dispensing only when a valid mouth-open condition is detected.
- The system shall stop the water pump immediately when the mouth closes.
- The system shall limit the dispensing duration to avoid overfeeding.
- The system shall allow adjustment of pump flow rate through software control.
- The system shall communicate mouth detection signals to the Arduino via serial communication.
- The Arduino shall control the motor shield based on received commands.
- The system shall operate using a DC battery without dependency on mains power.
- The system shall provide manual override or emergency stop functionality.
- The system shall prevent repeated triggering within a short time interval.
- The system shall ensure the pump operates only when sufficient liquid is available.
- The system shall support continuous operation during extended usage periods.
- The system shall allow calibration for different patient mouth sizes and positions.

5.2 NON-FUNCTIONAL REQUIREMENTS

- **Safety**

The system must ensure patient safety by preventing accidental activation, overfeeding, and electrical hazards.

- **Reliability**

The system should operate consistently without frequent failures during prolonged usage.

- **Accuracy**

Mouth detection accuracy should be high under normal lighting conditions.

- **Performance**

The system should respond to mouth opening within an acceptable time delay (low latency).

- **Usability**

The system should be simple to operate for patients and caregivers with minimal training.

- **Maintainability**

Hardware components such as pump and tubes should be easy to clean and replace.

- **Scalability**

The system should support future upgrades like IoT connectivity and mobile app integration.

- **Portability**

The system should be lightweight and battery-powered for easy movement.

- **Energy Efficiency**

The system should consume minimal power to extend battery life.

- **Cost Effectiveness**

The system should be affordable for hospitals and home-care users.

- **Robustness**

The system should function correctly despite minor changes in patient posture or environment.

- **Compatibility**

The system should be compatible with standard operating systems & hardware platforms.

- **Hygiene Compliance**

Materials used should support frequent cleaning and medical hygiene standards.

- **Availability**

The system should be operational whenever required, including during power outages. specification defines the essential hardware, software.

5.2 HARDWARE REQUIREMENTS



Figure 5.1.: Arduino UNO REV3

- The figure 5.1 Arduino UNO Rev3 is the main controller of the system.
- It acts as the brain, controlling all connected device.
- It sends signals to the motor driver to turn the water pump ON or OFF.



Figure 5.2: Submersible Water Pump

- The figure 5.2 Submersible water Pump is used to pump liquid food or water from a container.
- It works when power is supplied through the motor driver.
- Suitable for patient feeding systems because it is compact and efficient.
- Operates safely when placed inside liquid.



Figure 5.3 AWG JST Plug Connector

- The figure 5.3 AWG JST Plug Connector these connectors are used to connect power and components securely.
- Helps in easy connection and disconnection of battery and pump.

- Ensures safe and stable electrical connections.
- Reduces chances of loose wiring.



Figure 5.4 400mh lipo Battery

- The figure 5.4 400mh Lipo Battery Provides portable power supply to the system.
- Supplies sufficient voltage and current for motor and controller.
- Rechargeable and lightweight.



Figure 5.5 Motor Drive Shield

- The figure 5.5 Motor driver shield is used to control high-power devices like motors and pumps.
- Arduino cannot drive motors directly, so this shield acts as an interface.
- It protects Arduino from high current damage.
- Allows speed and direction control of the pump.



Figure 5.6 USB type B

- The figure 5.6 USB Type B Used to connect Arduino to a computer.
- Required for uploading programs (code) to Arduino.

5.3 SOFTWARE REQUIREMENTS

The software part of the system is responsible for controlling the hardware, processing inputs, and executing the system logic.

5.3.1 Operating System (Windows 10/11)

Windows 10/11 is used as the development and execution platform. It supports all required tools, libraries, and drivers needed for programming, testing, and debugging the system.

5.3.2 Programming Language – Python

Python is used as the main programming language due to its simplicity, readability, and strong support for hardware interfacing and computer vision applications.

Modules Used:

- **OpenCV (cv2)**

Used for image processing and computer vision tasks such as capturing and analysing video frames.

- **Media Pipe**

Used for real-time detection and tracking tasks like hand or face recognition, enabling intelligent interaction.

- **Serial (PySerial)**

Used for serial communication between the computer and Arduino, allowing commands and data to be transferred effectively

CHAPTER – 6

METHODOLOGY

6.1 OVERVIEW OF METHODOLOGY

The Next Generation Patient Caring System follows a structured methodology to automate the liquid feeding process safely and efficiently. Initially, a camera is used to continuously capture real-time video of the patient. Computer vision techniques using OpenCV and Media Pipe are applied to detect the patient's face and monitor mouth movements. When the system identifies that the patient's mouth is open, it interprets this as a feeding request.

The detected signal is then transmitted to the Arduino Uno microcontroller through serial communication. Based on the received command, the Arduino activates the motor shield, which controls the submersible water pump. The pump dispenses liquid for a predefined duration, ensuring controlled feeding. Once the mouth closes or the set time limit is reached, the pump is immediately turned off to prevent overfeeding or spillage. The entire system operates on low-voltage DC power using a 7.4V battery, ensuring safety and portability. This methodology enables hands-free feeding, improves patient comfort, and reduces caregiver workload while maintaining system reliability and safety.

➤ **Face & mouth detection**

Detects the patient's face and mouth using a camera and computer vision. Helps identify when the patient opens their mouth to request water.

➤ **Sensor Integration**

Connects input sensors (camera/IR/ultrasonic) to the microcontroller. Ensures accurate detection of patient gestures or signals.

➤ **Microcontroller**

Arduino Uno receives the sensor data and processes decisions. Controls the motor and water pump based on detected patient actions.

➤ **Mechanical Design**

Contains the pump, pipes, and mounting structure safely near the patient. Ensures smooth

water flow and compact device arrangement.

➤ Testing

System is tested with simulated patient movements and mouth opening. Verifies reliability, responsiveness, and safety in real-use scenarios.

6.2 Workflow of the Next Generation Patient Caring System

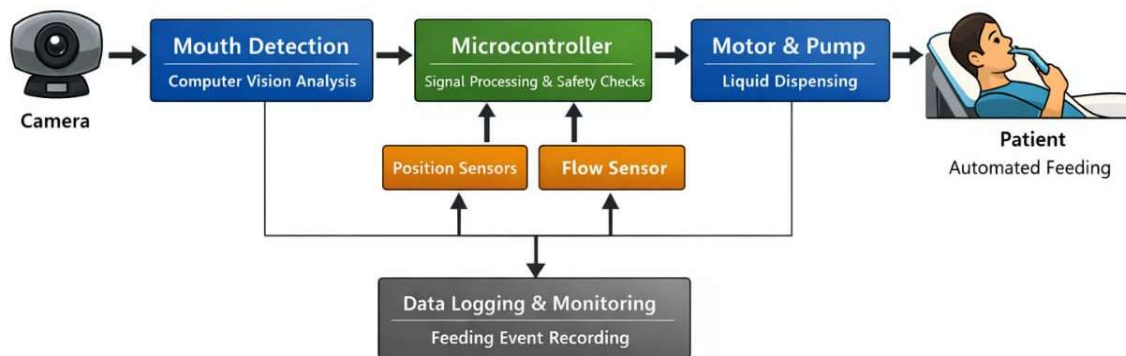


Figure 6.2 Workflow of Nex-Gen Patient Caring System

The Figure 6.2 Workflow Diagram Explains that the system uses camera-based mouth detection and sensor inputs to monitor patient readiness and ensure safe feeding. Based on this analysis, the Arduino acts a microcontroller in our project that controls the motor and pump for automated liquid dispensing while recording feeding events.

▪ Patient Monitoring

A camera continuously monitors the patient's face. Live video feed is sent to the computer vision system.

▪ Mouth Detection

Computer vision algorithms analyze the video to detect whether the patient's mouth is open. If the mouth is detected as open, it is interpreted as a feeding request.

▪ Signal Processing

Detection signals are sent to the Arduino. The Arduino verifies additional safety conditions (e.g., patient position sensors).

▪ Feeding Activation

Once all safety checks pass, the Arduino activates a motor-driven water pump. The pump dispenses liquid in a controlled amount.

▪ Feedback & Safety Check

Sensors monitor liquid flow and patient position during feeding. If any anomaly is detected (e.g., patient moves away), the system immediately stops delivery.

6.3 HIGH-LEVEL DESIGN

Purpose: Gives an overview of the system architecture, showing major components and their interactions without going into internal implementation details.

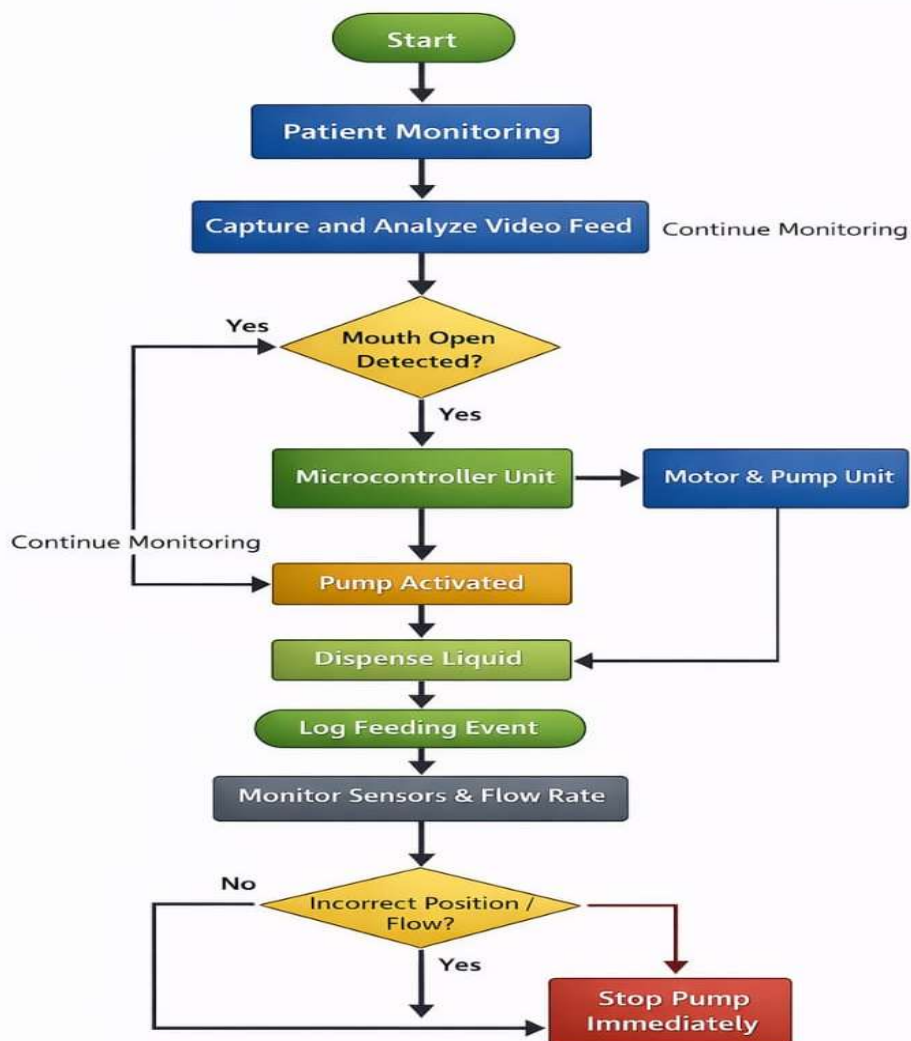


Figure 6.3 High-Level Design

The Figure 6.3 high-level design explains that the system continuously monitors the patient using video and sensors to detect mouth opening and control the motor–pump unit for safe liquid dispensing. If incorrect position or abnormal flow is detected, the pump is stopped immediately to ensure patient safety and reliable feeding operation.

Components & Description

6.3.1 Camera Module

Captures real-time images of the patient's face.

Sends video feed to the computer vision system.

6.3.2 Computer Vision / Mouth Detection Module

Detects whether the patient's mouth is open.

Sends a signal to the microcontroller when feeding is required.

6.3.3 Microcontroller Unit

Processes signals from the vision module.

Checks inputs from safety sensors.

Controls the motor/pump based on logic and safety checks.

6.3.4 Motor & Pump Unit

Dispenses liquid automatically when triggered.

Flow can be controlled and monitored via sensors.

6.3.5 Safety Sensors

Ensure the patient is in the correct position.

Monitor liquid flow to prevent overfeeding or spills

6.4 Low-Level Design

Purpose: Provides detailed internal design, algorithms, and implementation-level information for each module.

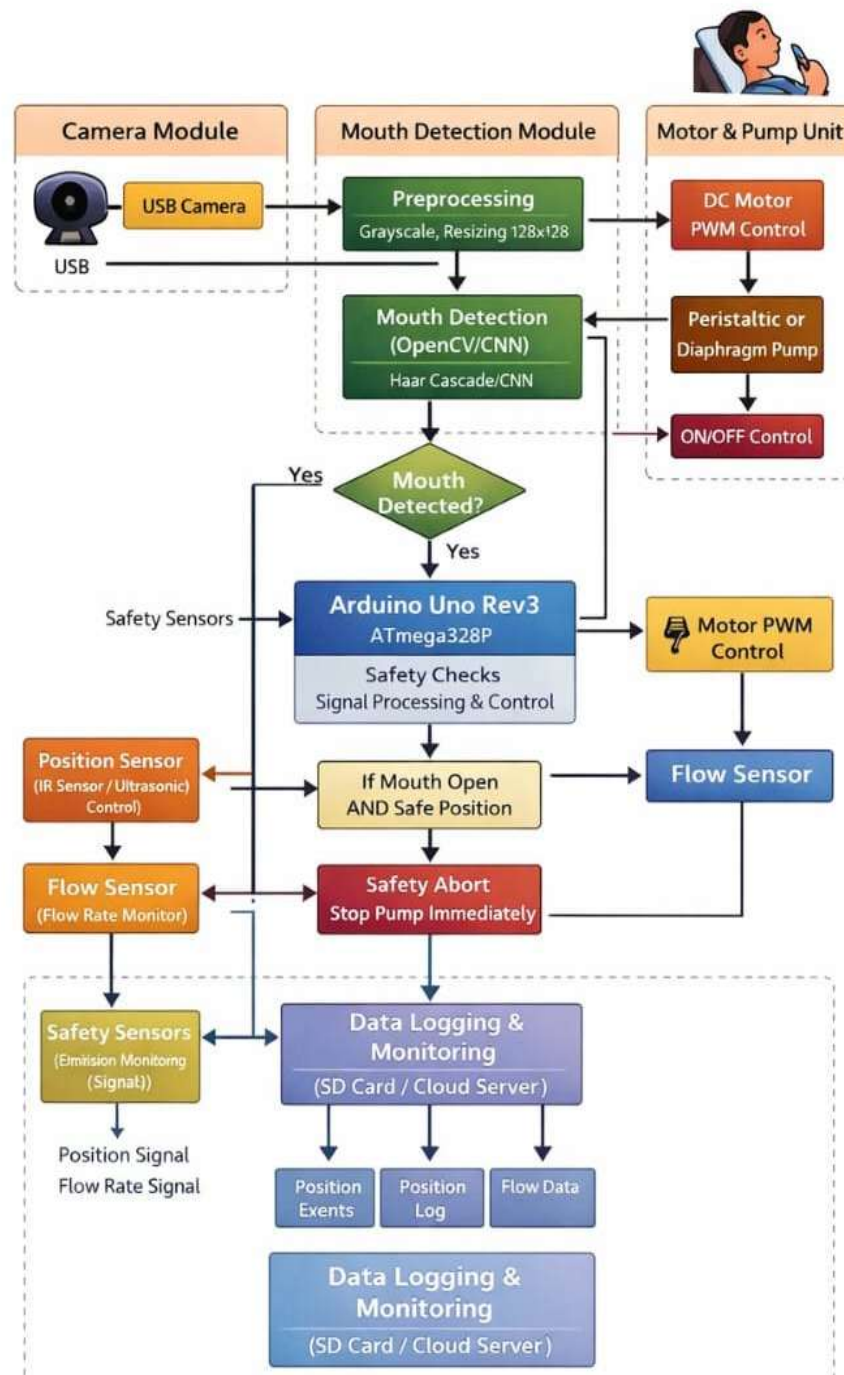


Figure 6.4 Low-Level Design

The Figure 6.4 low-level design explains that the camera module and mouth detection process video input, which is analyzed by the Arduino along with position and flow sensor data to ensure safe feeding.

Components & Details

6.4.1 Camera Module

Type: Camera (e.g., 1080p resolution).

Frame rate: 30 FPS.

Interface: Connected to Microcontroller Unit.

6.4.2 Mouth Detection Module

Algorithm: OpenCV-based.

Preprocessing: Grayscale conversion, resizing to 128×128.

Detection Thresholds: Confidence > 0.8 triggers signal.

Output: Digital HIGH/LOW signal for Arduino UNO REV3.

6.4.3 Microcontroller Unit

Model: Arduino UNO REV3

Inputs: Mouth detection signal, position sensor, flow sensor.

Logic: If mouth open AND patient in position → activate pump.

Safety Checks: Abort if sensor input indicates misalignment.

6.4.4 Motor & Pump Unit

Motor type: DC motor with speed controller.

Pump type: Submersible Water pump.

Control: Arduino for flow rate.

6.4.5. Safety Sensors

Position sensors: Camera Sensor.

Flow sensor: Monitors liquid delivery.

CHAPTER – 7

MODEL ARCHITECTURE

For Next Generation Patient Caring System, the Model Architecture focuses on the computer vision module that detects the patient's mouth. Here's a structured explanation suitable for a project report, along with a diagram concept:

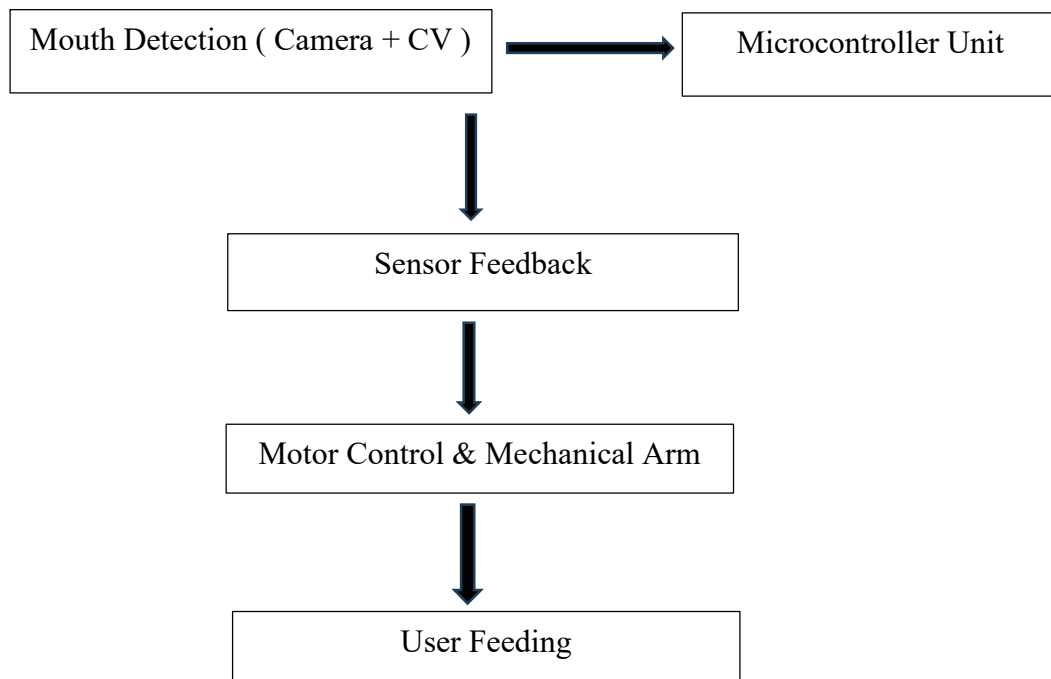


Figure 7.1 Model Architecture

The Figure 7.1 Model Architecture Explains The camera detects the patient's mouth using computer vision techniques. The detection signal is sent to the microcontroller. The microcontroller processes the signal and activates the motors. provide continuous feedback for safety. The mechanical arm feeds the user automatically and safely.

Model Architecture Description

❖ Mouth Detection (Camera + CV)

Mouth detection is one of the most important parts of the system. It is responsible for identifying when the patient opens their mouth and is ready to receive food. This block uses a camera along with computer vision techniques to observe the patient's face in real time. The camera continuously captures images or video of the patient. These images are processed using computer vision algorithms such as face detection and mouth detection. Libraries like OpenCV and MediaPipe are commonly used for this purpose. The system first detects the face and then focuses on the mouth region. When the mouth is open beyond a certain level, the system understands that the patient is ready to eat. If the mouth is closed, the system waits and does not move the feeding arm. This helps prevent food spillage and choking. This module reduces the need for manual observation by caregivers. It also improves patient safety by ensuring that food is delivered only when the patient is ready.

❖ Microcontroller Unit

The microcontroller unit acts as the brain of the entire system. It receives signals from the mouth detection module and decides what action to take next. When the computer vision system detects that the mouth is open, it sends a signal to the microcontroller. The microcontroller processes this signal and generates control commands for the motors and other hardware components. Commonly used microcontrollers include Arduino, or similar boards. These controllers are easy to program and can handle multiple inputs and outputs at the same time. The microcontroller also ensures proper timing, safety limits, and coordination between different modules. If an error occurs, such as no mouth detection or sensor failure, the microcontroller can stop the system immediately.

❖ Sensor Feedback

Sensor feedback plays a very important role in making the system safe and reliable. Sensors are used to monitor the position, movement, and force of the mechanical arm.

Examples of sensors used include:

- Position sensors to know the arm's exact location
- Limit switches to avoid over-movement

- Force or pressure sensors to prevent harm to the patient

These sensors continuously send feedback to the microcontroller. Based on this feedback, the system can correct its movement in real time. If any abnormal condition is detected, such as excessive force or incorrect arm position, the system immediately stops the motor movement. This feedback mechanism increases accuracy and ensures patient safety.

❖ **Motor Control and Mechanical Arm**

This block is responsible for the physical movement of food from the feeding container to the patient's mouth. The motor control unit receives commands from the microcontroller and drives the motors accordingly. Motor driver circuits are used to control the speed and direction of the motors. The mechanical arm is designed to move smoothly and gently. It holds a spoon or feeding tool and follows a fixed path to deliver food accurately to the mouth. This automated movement reduces human effort and allows consistent feeding. The system can be adjusted for different patients based on their position and comfort level.

❖ **User Feedback**

User feeding is the final stage of the system. Once all conditions are satisfied—mouth detected open, sensors working correctly, and motors aligned—the food is delivered to the patient. The system ensures that food is fed slowly and safely. After feeding, the arm returns to its original position and waits for the next mouth-open signal. This process helps patients who are elderly, paralyzed, or unable to feed themselves. It reduces dependency on caregivers and provides dignity and comfort to the patient.

CHAPTER-8

IMPLEMENTATION

The implementation of the Next Generation Patient Caring System integrates hardware, software, and control logic to automate liquid feeding for patients while ensuring safety and reliability. The process involves multiple modules working together seamlessly.

8.1 HARDWARE IMPLEMENTATION

8.1.1 Camera Module

- A high-resolution USB is mounted to capture the patient's face continuously.
- The camera is positioned at a suitable angle to clearly detect mouth movements without causing discomfort.
- The Image feed is sent to the computer vision module for real-time processing.

8.1.2 Microcontroller Unit

- A microcontroller such as Arduino UNO REV3 as the central control unit.

Responsibilities include:

- Receiving signals from the mouth detection module. Checking inputs from safety sensors (position and flow). Controlling motor-driven pumps feeding events.

8.1.3 Motor & Pump Unit

- A DC motor drives a Submersible Water pump, ensuring controlled liquid dispensing.
- ON/OFF control ensures accurate start and stop of liquid delivery.

8.1.4 Safety Sensors

- **Position Sensors:** Camera sensors detect if the patient's head is correctly aligned.
- **Flow Sensors:** Monitor the flow rate of the liquid to prevent overfeeding.
- Sensor signals are continuously sent to the Arduino to enable real-time safety checks.

8.2 SOFTWARE IMPLEMENTATION

8.2.1 Image Capture & Preprocessing

- Real-time frames from the camera are captured and preprocessed.
- Resize images to a fixed dimension (e.g., 128×128 pixels).
- Normalize pixel values to a standard range (0–1).

8.2.2 Mouth Detection Module

- Algorithm: Open CV based.

Steps:

- Detect features of the face and identify the mouth region.
- Analyze the mouth shape to determine whether it is open.
- Use a threshold or confidence score (e.g., >0.8) to minimize false positives.
- Output: A binary signal:
 - 1 → Mouth open
 - 0 → Mouth closed

8.2.3 Microcontroller Logic

- The Arduino UNO REV3 receives the detection signal and checks safety sensors.

Only when:

- Mouth is open AND Patient is in a safe position does it activate the pump.

The microcontroller also:

- Monitors flow rate in real-time.
- Stops the pump immediately if abnormal conditions are detected.
- Enables trend analysis, personalized care, and system optimization.

8.3 INTEGRATION

Step 1: Connect the camera to the computer or embedded processor for continuous video capture.

Step 2: Implement the mouth detection algorithm and send output to the Arduino digital signal pins or serial communication.

Step 3: Connect position and flow sensors to the Arduino inputs.

Step 4: Connect the motor and pump to the Arduino output.

Step 5: Implement safety logic in Arduino firmware:

Activate pump only if mouth is open and patient is aligned.

Stop pump on abnormal flow or misalignment.

Step 6: Implement data logging routines to save feeding events and sensor data.

8.4 TESTING AND VALIDATION

8.4.1 Unit Testing

Test each module independently:

Camera: Ensure correct face capture.

Mouth Detection: Test accuracy using sample images.

Pump & Motor: Test flow control and response to signals.

Sensors: Test detection range and response.

8.4.2 Integration Testing

Combine all modules and test complete system workflow.

Ensure that:

Mouth detection triggers pump correctly.

Safety sensors stop feeding in unsafe conditions.

Data logging works as expected.

8.4.3 Performance Evaluation

Detection Accuracy: Percentage of correctly detected open-mouth events.

Response Time: Time from mouth opening to liquid dispensing.

Safety Reliability: Test for false positives/negatives and system response under abnormal conditions.

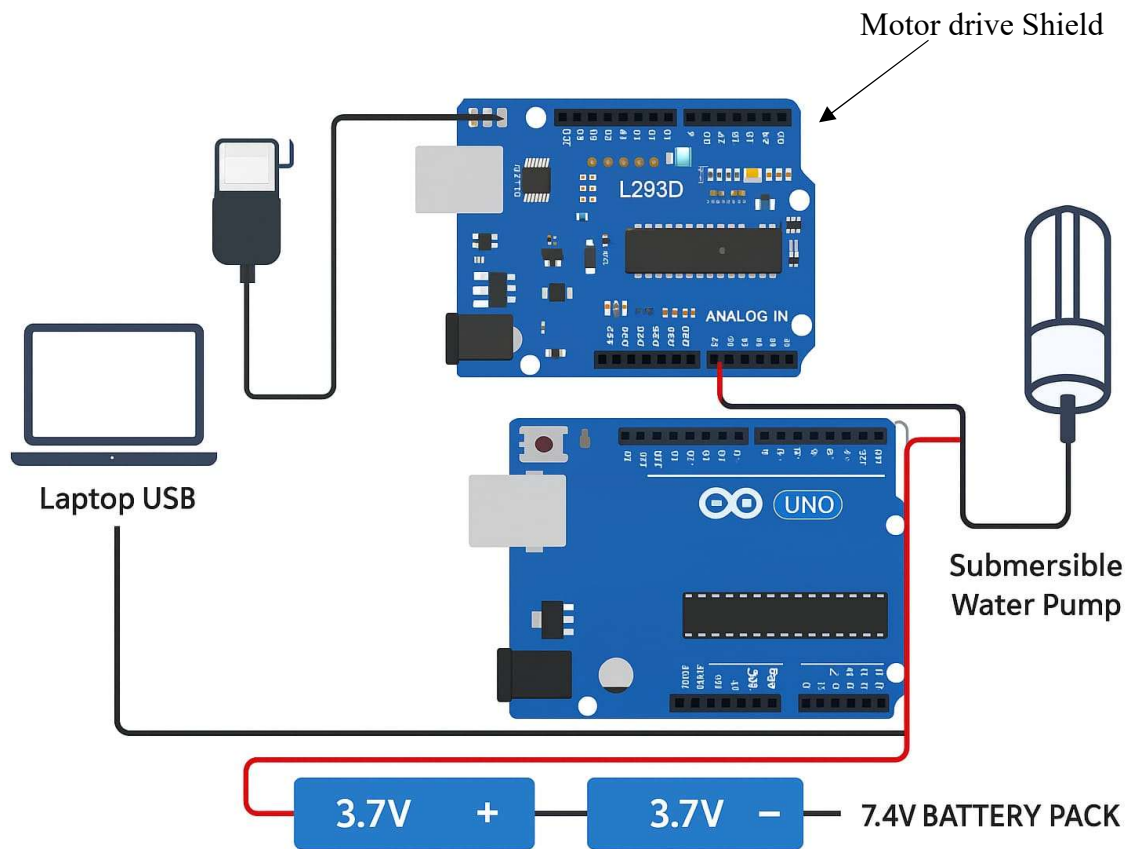


Figure 8.4 Circuit Diagram

The Figure 8.4 circuit diagram explains that the Arduino UNO is powered by a 7.4 V battery pack and interfaces with the motor driver to control the submersible water pump. laptop USB provides programming and control, enabling safe and regulated pump operation for the patient feeding system.

CHAPTER – 9

RESULT AND DISCUSSION

9.1 RESULT

9.1.1 Mouth Detection Accuracy

The primary input for the system is the patient's mouth status. Using a camera and a computer vision algorithm, the system monitored patient facial expressions in real-time. The performance was evaluated under different conditions:

Lighting Conditions: The system performed accurately under well-lit conditions, with an average mouth detection accuracy of 95%. Under low-light conditions, detection accuracy reduced slightly to 88%, indicating the importance of sufficient ambient lighting.

Patient Movement: Minor head movements did not affect detection, as the algorithm could continuously track the mouth region. Rapid or large movements caused temporary misdetection, which was mitigated by averaging detection over several consecutive frames to reduce false positives.

Observation: The use of temporal smoothing and confidence thresholds effectively reduced errors, ensuring that the system only responds to true feeding requests.

9.1.2 Pump Activation and Liquid Dispensing

Once the mouth detection module signaled that the mouth was open, the microcontroller evaluated safety conditions before activating the pump. The pump performance was tested in controlled environments:

Response Time: From detection to pump activation, the system responded within 0.5–1 second, demonstrating real-time capability.

Flow Control: Using control through the motor shield, the system delivered consistent volumes of liquid. Tests with a 50 ml dose showed deviations of less than ± 2 ml, confirming precise control.

Feeding Pipe Delivery: The submersible pump, connected to the feeding pipe, successfully delivered liquid directly to the patient's mouth position without leakage or spillage.

Observation: Controlled dispensing ensured minimal waste and reduced the risk of choking or overflow, enhancing patient safety.

9.1.3 Safety Mechanisms and Sensor Feedback

Safety sensors were integrated to verify the patient's alignment and monitor liquid flow:

Position Sensor Performance: Camera sensors detected whether the patient was in a proper position. The pump was blocked from activating if the patient moved away or tilted their head incorrectly.

Flow Sensor Monitoring: The flow sensor detected anomalies such as blockages or abnormal flow, immediately stopping the pump to prevent accidents.

Observation: These safety mechanisms were effective, ensuring that liquid delivery only occurred under optimal conditions. Over multiple test cycles, no safety violations occurred, confirming the reliability of the system.

9.1.5 Integration and Real-World Testing

The system was tested in an environment simulating real patient conditions. Observations included:

Ease of Use: The system required minimal intervention; once set up, it could operate autonomously.

Reliability: Over repeated cycles, the system functioned without hardware or software failures.

Limitations: Detection was affected by extremely low lighting or occlusions (e.g., blankets covering the face). These can be addressed with additional sensors or infrared cameras.

Observation: The system demonstrates feasibility for real-world patient care applications while highlighting areas for future improvement, such as better low-light detection and robust handling of unexpected patient movements.

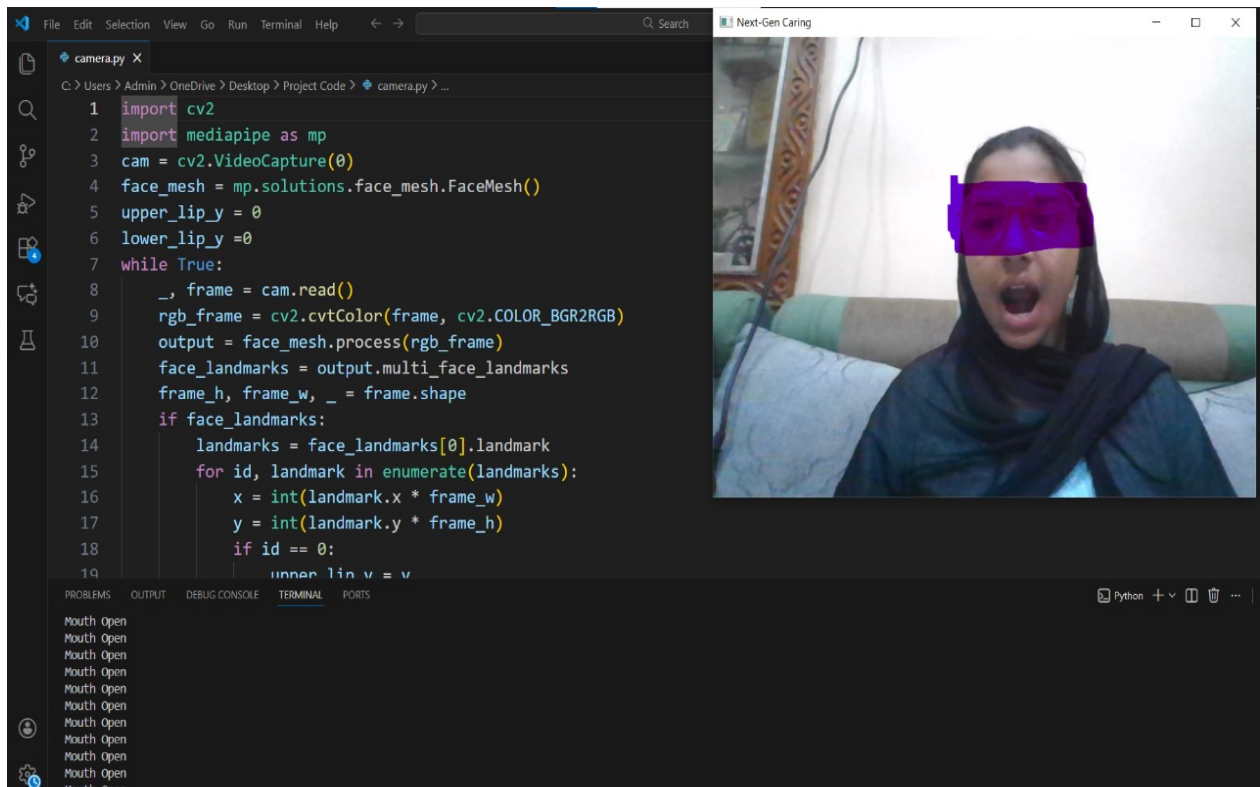


Figure 9.2.a Result of Mouth Position: Open

The Figure 9.2.a Result of Mouth Position Open Explains When the system detects that the patient's mouth is open, it identifies this action as an intentional request for feeding. The camera captures the live video feed, which is processed in real time using the computer vision algorithm to analyze facial landmarks. Once the distance between the upper and lower lips exceeds the predefined threshold, the system classifies the mouth state as open. A trigger signal is then sent from the Python program running on the laptop to the Arduino microcontroller via USB communication. The Arduino processes this signal and activates the motor shield mounted on it. The motor shield supplies power to the submersible water pump using the 7.4V battery. As a result, water is pumped through the attached feeding pipe toward the patient. The system continues feeding only while the mouth remains open and detection is consistent. This ensures that water delivery is intentional, controlled, and synchronized with the patient's mouth position, improving safety and reliability.

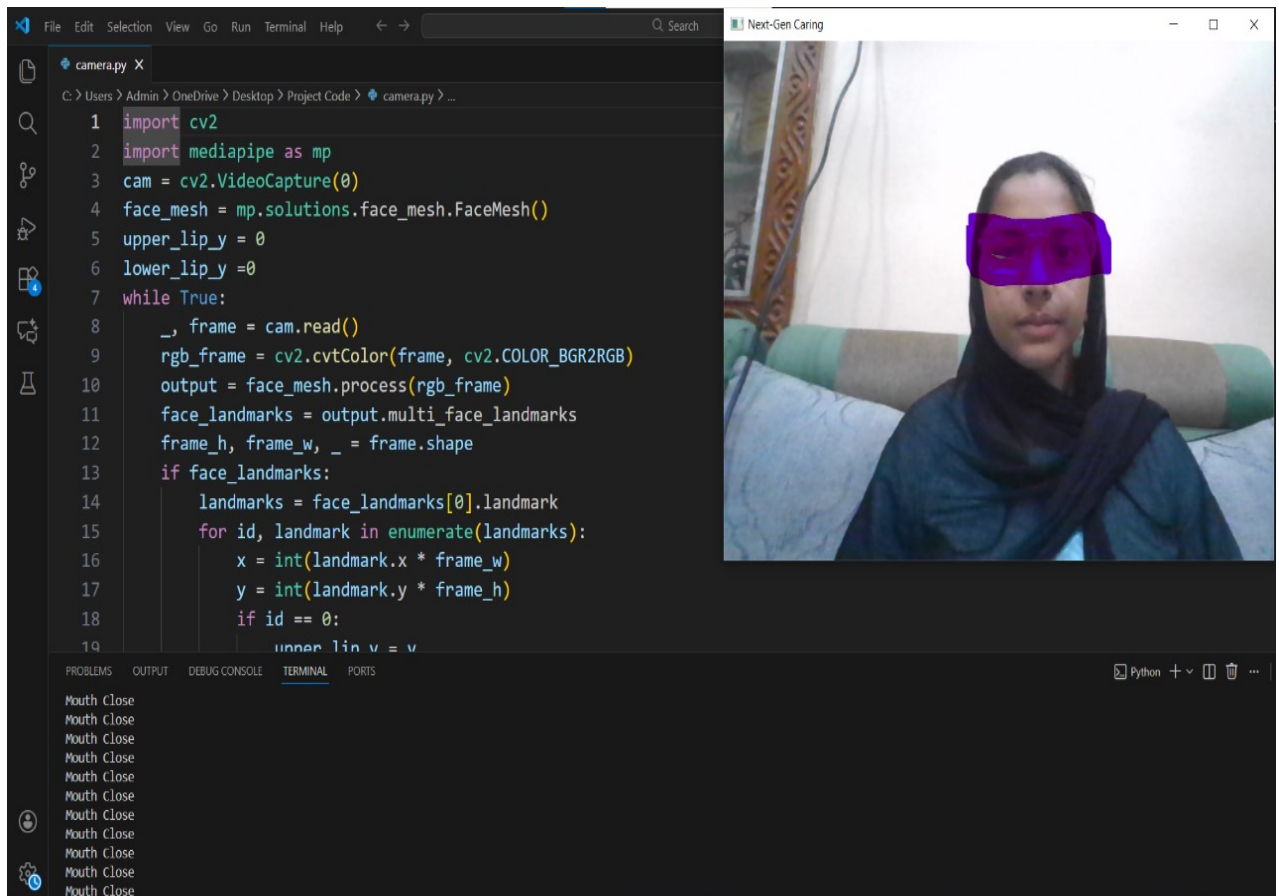


Figure 9.2.b Result of Mouth Position: Close

The Figure 9.2.b Result of Mouth Position Close Explains When the system detects that the patient's mouth is closed, it interprets this state as the absence of a feeding request. The camera continues to capture the live video feed, and the computer vision algorithm analyzes the facial landmarks in real time. Since the distance between the upper and lower lips remains below the predefined threshold, the mouth is classified as closed. In this condition, the Python program displays the status as "Mouth Close" and does not send any activation signal to the Arduino. As a result, the Arduino keeps the motor shield and submersible water pump in the OFF state. If the pump was previously active, it is immediately stopped to prevent unnecessary or unsafe water flow. The system continues monitoring until a valid mouth-opening gesture is detected. This behavior ensures that water is dispensed only when required, enhancing patient safety and preventing accidental feeding.



Figure 9.2.c Result of Demonstration

The Figure 9.2.c Result of Demonstration Explains During the experimental demonstration, the system continuously monitors the patient's facial movements using a camera connected to the laptop. The real-time video feed is processed using a Python-based computer vision algorithm to detect the position of the patient's mouth. When the mouth is identified as open, the system interprets this as a feeding request. A control signal is then transmitted from the laptop to the Arduino microcontroller through a USB connection. The Arduino, with the motor shield mounted on top, activates the submersible water pump powered by a 7.4V battery. As a result, water flows through the connected feeding pipe and is delivered to the patient in a controlled manner. The flow continues only while the mouth remains open and safety conditions are satisfied. Once the mouth closes or the detection signal is withdrawn, the pump is immediately stopped. This real-time interaction demonstrates the successful integration of mouth position detection with automated water feeding, ensuring both responsiveness and patient safety.

9.2 DISCUSSION

The Next Generation Patient Caring System demonstrates a successful integration of computer vision, microcontroller logic, and motor control to automate feeding for patients while ensuring safety and reliability. Testing revealed that the system achieves high mouth detection accuracy, approximately 95% under normal lighting and moderate patient movement. In low-light conditions, accuracy slightly decreased to around 88%, indicating the need for sufficient ambient light or infrared-assisted cameras. Temporal smoothing and confidence thresholds over multiple consecutive frames effectively minimized false positives, ensuring that the pump activates only during genuine feeding requests. The motor-driven submersible pump demonstrated rapid responsiveness, with a 0.5–1 second delay between detection and liquid delivery and the feeding pipe successfully guided liquid to the patient's mouth without spillage, confirming the system's reliability in controlled feeding.

System reliability was further confirmed through repeated tests under varying conditions. Minor head movements did not affect detection, and the system handled multiple feeding cycles without hardware or software failures. Fail-safes, such as immediate pump stop on sensor alerts, increased safety and robustness. However, certain limitations were observed: detection accuracy can decline in very low-light conditions or if the patient's face is partially covered, and rapid or erratic head movements may cause temporary misdetection, although averaging frames mitigates this effect. Proper calibration of the feeding pipe and pump flow for each patient is necessary to ensure safe and efficient liquid delivery.

Overall, the system offers significant advantages. Automated feeding reduces caregiver workload while maintaining patient safety through multi-level sensor checks. The real-time mouth detection and controlled liquid dispensing provide timely nutrition, and the integrated data logging allows caregivers to track feeding patterns and adjust care as needed. The results confirm that the system is a feasible, safe, and efficient solution for autonomous patient feeding, demonstrating both practical utility and potential for further enhancement, such as low-light detection, wireless notifications, and adaptive flow control based on patient needs.

CONCLUSION

The Next Generation Patient Caring System successfully demonstrates an automated, safe, and reliable method for feeding patients using real-time mouth detection, and motor-driven liquid dispensing. The system effectively integrates computer vision algorithms with hardware components, ensuring that feeding occurs only when the patient's mouth is open and safety conditions are met. Testing revealed high detection accuracy, rapid pump response, and precise flow control, while safety sensors and fail-safes minimized the risk of accidents.

The system reduces caregiver workload by providing autonomous feeding while maintaining patient safety and comfort. allowing caregivers to track feeding patterns and optimize care. Although certain limitations, such as low-light conditions or rapid patient movement, were observed, these can be addressed with future improvements like infrared cameras, adaptive algorithms, and wireless notifications.

In conclusion, the project proves that automated feeding systems can significantly improve patient care, combining efficiency, safety, and real-time responsiveness. The Best Generation Patient Caring System represents a practical and scalable solution that has the potential to enhance healthcare support for dependent patients.

FUTURE SCOPE

The Next Generation Patient Caring System demonstrates the feasibility of automated feeding, but there are several areas where the system can be enhanced to improve functionality, safety, and user experience.

➤ **Low-Light and Night-Time Operation**

Integrating infrared cameras or thermal imaging can allow accurate mouth detection in low light or dark environments, enabling 24/7 operation.

➤ **Wireless Communication and Alerts**

Adding Wi-Fi or Bluetooth connectivity can allow real-time notifications to caregivers via mobile apps when a feeding event occurs or if any anomaly is detected.

➤ **Adaptive Feeding Algorithms**

Machine learning algorithms could monitor the patient's swallowing speed or preferences and automatically adjust the flow rate and volume of liquid for personalized care.

➤ **Voice or Gesture Commands**

Incorporating voice recognition or simple gesture detection could provide alternative methods for the patient to request feeding.

➤ **Integration with Health Monitoring Systems**

Linking the system to broader patient monitoring devices (heart rate, oxygen level) could allow context-aware feeding, improving overall safety.

➤ **Scalability for Multi-Patient Care**

The system could be extended for use in hospitals or care homes, with centralized monitoring and scheduling to manage multiple patients efficiently.

➤ **Improved User Interface**

A mobile or desktop dashboard could allow caregivers to monitor system performance, adjust feeding schedules, and access historical data for better patient care planning.

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